

DBF Compatibility Matrix – Harbour 3.2.0dev

Revised & Extended Technical Reference | Includes CA-Clipper, Nantucket Clipper, Paradox, NSX and all native RDDs | Harbour GitHub HEAD (2025)

NATIVE	Full support: reliable read, write and index maintenance via Harbour native RDD.
PARTIAL	Reading generally possible; index creation/maintenance limited or RDD-dependent.
NOT SUPP.	Proprietary/incompatible format. Access only via external ODBC/ADO/BDE.
EXTERNAL	Support via external RDD/library not included in standard Harbour.

1. DBF Format Matrix by Header Byte

† CA-Clipper and Nantucket Clipper both use header byte 0x03 (identical to dBASE III+); distinguished solely by index RDD (.NTX). ‡ Paradox does NOT use DBF format — .DB is proprietary and incompatible with any xBase RDD.

Dec	Hex	Memo	Indexes	Type	Support	RDD	Origin / Description	Technical Notes
2	0x02	—	.NDX/.NTX	Single	NATIVE	DBFNTX	FoxBASE original	No memo. Compatible with Clipper and CA-Clipper.
3	0x03	—	.NDX/.NTX	Single	NATIVE	DBFNTX	dBASE III Plus / FoxBASE+ (no memo)	Most common legacy Clipper format. Full read/write/index support.
48	0x30	.FPT	.CDX	Compound	NATIVE	DBFCDX	Visual FoxPro (standard)	Full support via DBFCDX. FPT memo VFP-compatible.
49	0x31	.FPT	.CDX	Compound	NATIVE	DBFCDX	Visual FoxPro (autoincrement)	Autoincrement field read OK; write needs attention.
50	0x32	.FPT	.CDX	Compound	PARTIAL	DBFCDX	Visual FoxPro (VARCHAR/VARBINARY)	VARCHAR/VARBINARY read as Character. No variable-length native support.
67	0x43	—	.MDX	Compound	PARTIAL	DBFMdx*	dBASE IV SQL table (no memo)	DBFMdx available but limited. MDX not a native Harbour RDD.
99	0x63	—	.MDX	Compound	PARTIAL	DBFMdx*	dBASE IV SQL system files	Partial read only. Avoid in production.
131	0x83	.DBT	.NDX/.NTX	Single	NATIVE	DBFNTX	dBASE III Plus / FoxBASE+ (with memo)	DBT memo dBASE III. Full support. Typical Clipper migration format.
139	0x8B	.DBT	.MDX	Compound	PARTIAL	DBFMdx*	dBASE IV (with memo)	DBT memo dBASE IV. MDX partial support only.
203	0xCB	.DBT	.MDX	Compound	PARTIAL	DBFMdx*	dBASE IV SQL table (with memo)	Same as 0x8B with SQL flag. Rare in practice.
229	0xE5	.SMT	.NTX	Single	PARTIAL	SIXRDD*	HiPer-Six / Six Driver	Proprietary SMT memo. Requires SIXRDD. Avoid in new apps.
245	0xF5	.FPT	.CDX/.IDX	Compound	NATIVE	DBFCDX	FoxPro 2.x (with memo)	FoxPro 2.x FPT memo. Fully compatible via DBFCDX.
251	0xFB	—	.NDX/.NTX	Single	NATIVE	DBFNTX	FoxBASE (variant)	FoxBASE variant. No memo. Identical to 0x02/0x03.
3†	0x03	—	.NTX	Single	NATIVE	DBFNTX	Nantucket Clipper / CA-Clipper	Uses 0x03 with exclusive .NTX. DBFNTX is Harbour's native RDD for this.
—	—	—	.NSX	Compound	NATIVE	DBFNSX	CA-Clipper 5.x / Harbour NSX	Harbour native compound index. Recommended over CDX in pure Harbour environments.

Dec	Hex	Memo	Indexes	Type	Support	RDD	Origin / Description	Technical Notes
3‡	0x03	.DBT	.DB/—	—	NOT SUPP.	—	Borland Paradox	NOT DBF. Paradox .DB is proprietary. Access only via ODBC/BDE/ADO.
—	—	.DBT	.NTX/.CDX	S/Comp	NATIVE	NTX/CDX	Harbour native DBF (self-generated)	Harbour outputs 0x03 or 0x30 by default, controlled by active RDD.

2. Index Format Compatibility

Ext.	Type	Support	RDD	Origin	Features / Limitations
.NTX	Single	NATIVE	DBFNTX	Nantucket Clipper / CA-Clipper / Harbour	Native Harbour RDD. One key per file. Full DESCEND, FOR, UNIQUE support.
.CDX	Compound	NATIVE	DBFCDX	FoxPro / Visual FoxPro / Harbour	Multi-tag compound index. Fully supported. VFP CDX compatible.
.NSX	Compound	NATIVE	DBFNSX	Harbour native (modern)	Harbour's own compound index. Recommended for new apps. Better perf than NTX.
.NDX	Single	NATIVE	DBFNDX	dBASE III / FoxBASE	Oldest format. Supported via DBFNDX. Functional but avoid in new projects.
.MDX	Compound	PARTIAL	DBFMdx*	dBASE IV / dBASE 5	Partial support. DBFMdx not actively maintained. Test before use.
.IDX	Single	NATIVE	DBFCDX	FoxPro 2.x	Simple FoxPro index. DBFCDX handles read/write correctly.
.DB	—	NOT SUPP.	—	Borland Paradox	NOT a DBF index. Proprietary Paradox format. ODBC/BDE access only.
.BAG	Compound	PARTIAL	SIXRDD*	HiPer-Six / Six Driver	HiPer-Six compound index. Requires external SIXRDD.

3. Memo File Compatibility

Ext.	Support	Origin	Harbour Compatibility
.DBT	NATIVE	dBASE III / dBASE IV / Clipper / CA-Clipper	Most common legacy memo. Full support via DBFNTX and DBFCDX.
.FPT	NATIVE	FoxPro 2.x / Visual FoxPro	Fully supported by DBFCDX. Handles MEMO, GENERAL and PICTURE fields.
.SMT	PARTIAL	HiPer-Six	Proprietary HiPer-Six memo. Partial read via SIXRDD. Avoid.
.DBV	NOT SUPP.	dBASE 5 for Windows	dBASE 5 memo. Not natively supported. ODBC access only.

4. Harbour RDD Summary – Selection Guide

RDD	Status	DBF Byte	Index	Memo	Recommended Usage
DBFNTX	NATIVE	0x02/0x03/0x83	.NTX	.DBT	Legacy Clipper/CA-Clipper systems. Maintenance and migration.
DBFCDX	NATIVE	0x30/0xF5	.CDX/.IDX	.FPT	FoxPro/VFP interop. New apps needing compound indexes. Strongly recommended.
DBFNSX	NATIVE	0x03	.NSX	.DBT/.FPT	Pure Harbour native. Best choice for new Harbour applications.
DBFNDX	NATIVE	0x02/0x03	.NDX	.DBT	dBASE III/FoxBASE compatibility. Avoid in modern production code.
DBFMDX	PARTIAL	0x43/0x63/0x8B	.MDX	.DBT	dBASE IV MDX. Partial support. Read legacy files only, avoid index creation.
SIXRDD	EXTERNAL	0xE5	.BAG/.NTX	.SMT	External HiPer-Six library. Not bundled with Harbour. Legacy-specific only.
LETO	NATIVE	any	any	any	Remote DBF over network (Ieto server). Works on top of DBFNTX/CDX/NSX.
SQLRDD	NATIVE	—	—	—	SQL access (MySQL, PostgreSQL, SQLite, MSSQL) using xBase syntax.

5. Important Technical Notes

- **CA-Clipper / Nantucket Clipper:** Both use header byte 0x03 (identical to dBASE III+). The distinction is made by the index: Clipper uses exclusively .NTX files. Harbour's DBFNTX RDD was designed for maximum compatibility with this format.
- **Paradox (Borland):** NOT a DBF format. Paradox .DB files have a completely different binary structure and cannot be opened by any xBase RDD. Access exclusively via ODBC (Paradox ODBC driver), BDE, or ADO. In Harbour: win_oleCreateObject('ADODB.Connection') with MSDASQL provider.
- **NSX vs. CDX vs. NTX:** For new pure-Harbour applications prefer DBFNSX (native compound index, best performance). For FoxPro/VFP interoperability use DBFCDX. For legacy Clipper compatibility use DBFNTX.
- **MDX (dBASE IV):** DBFMDX RDD exists in Harbour but is not actively maintained. For dBASE IV migration, convert indexes to CDX or NSX.
- **Runtime RDD Selection:** Use REQUEST DBFNTX / DBFCDX / DBFNSX at program start and rdbSetDefault('DBFCDX') to set the global default. Different tables may use different RDDs simultaneously via USE <file> VIA 'DBFNTX'.
- **Empirical Validation:** This matrix is a technical reference. Always validate with representative datasets in a staging environment before production migration.

* RDD marked with asterisk requires additional library/component not included in standard Harbour. Technical reference only — does not replace empirical testing. Harbour 3.2.0dev | MinGW / Windows | GTWVT | 2025